Quiz — 1.1.2 Types of Processor

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.1.2

Section A — Multiple choice (8 marks)

1. A key feature of the **Von Neumann** architecture is that it uses: A. Separate memories and buses for data and instructions B. A single shared memory and bus for both data and instructions C. Thousands of small cores for parallel processing D. Only read-only memory Your answer: ☐

2. Which architecture allows an instruction to be **fetched at the same time** as data is read or written? A. Von Neumann B. CISC C. Harvard D. Single-core Your answer: ☐

3. Compared with CISC, **RISC** processors typically have: A. Many complex, variable-length instructions B. Few simple, fixed-length instructions aiming for one cycle each C. All complexity built into the hardware D. Higher power consumption Your answer: ☐

4. In a **CISC** processor, the complexity of executing instructions is mainly handled by the: A. Compiler / software B. Hardware C. Cache D. Address bus Your answer: ☐

5. Which type of processor is most commonly found in **mobile phones and embedded devices** because of its low power use? A. CISC (x86) B. RISC (ARM) C. Von Neumann only D. GPU only Your answer: ☐

6. A **GPU** is best suited to: A. Highly sequential, branch-heavy tasks B. Applying the same operation to large blocks of data in parallel C. Running the operating system kernel D. Storing data permanently when the power is off Your answer: ☐

7. Which of the following is an example of **GPGPU** (general-purpose computing on a GPU)? A. Rendering pixels for a 3D game B. Training a neural network / machine learning C. Decoding a single instruction in the CIR D. Holding the BIOS firmware Your answer: ☐

8. A task will **not** benefit much from parallel processing when: A. It can be split into many independent sub-tasks B. Each step depends on the result of the previous step (data dependency) C. The software is written to use multiple threads D. There are many cores available Your answer: ☐

Section B — Short answer (12 marks)

9. State **three** differences between Von Neumann and Harvard architectures. **[3]**

10. Many modern CPUs are described as a **modified (hybrid) Harvard** architecture. Explain what this means. **[2]**

11. Explain **two** differences between CISC and RISC processors. **[2]**

12. Explain why a **GPU** can render 3D graphics so quickly. **[2]**

13. Define the term **multicore processor**. **[1]**

14. A program sorts a list and then searches it. The search cannot start until the sort has finished. Explain why splitting this work across multiple cores gives **limited** benefit. **[2]**

Section C — Exam-style question (5 marks)

15. A research team needs to run large-scale scientific simulations that apply the same mathematical calculation to millions of data points. They are deciding whether to invest in a powerful multicore **CPU** or a high-end **GPU**.

Discuss which processor type is better suited to this workload and justify your answer. **(AO2/AO3) [5]**